

"""

Question 1: PCA on Customer Purchase Behavior Dataset

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"""

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.impute import SimpleImputer

# -----
# 1. Generate Synthetic Dataset
# -----

np.random.seed(42)

n = 300

data = {
    'CustomerID': range(1, n+1),
    'Annual_Income': np.random.normal(60000, 20000, n),
    'Spending_Score': np.random.randint(1, 101, n),
    'Age': np.random.randint(18, 70, n),
    'Frequency': np.random.poisson(12, n),
    'Avg_Order_Value': np.random.normal(150, 50, n),
    'Electronics': np.random.randint(0, 2, n),
    'Clothing': np.random.randint(0, 2, n),
    'Groceries': np.random.randint(0, 2, n),
    'Home_Decor': np.random.randint(0, 2, n),
    'Loyalty_Score': np.random.uniform(1, 10, n),
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Introduce 5% missing values
```

```
for col in ['Annual_Income', 'Spending_Score', 'Avg_Order_Value']:
```

```
    idx = np.random.choice(df.index, size=int(0.05*n), replace=False)
```

```
    df.loc[idx, col] = np.nan
```

```
print("=" * 55)
```

```
print(" QUESTION 1: PCA – Customer Purchase Behavior")
```

```
print("=" * 55)
```

```
print(f"\nDataset Shape : {df.shape}")
```

```
print(f"\nMissing Values:\n{df.isnull().sum()}")
```

```
# _____
```

```
# 2. Preprocessing
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# _____
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```
features = [c for c in df.columns if c != 'CustomerID']
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```
X = df[features].copy()
```

```
# Impute missing values with column mean
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```
imputer = SimpleImputer(strategy='mean')
```

```
X_imputed = pd.DataFrame(imputer.fit_transform(X), columns=features)
```

```
# Normalise
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X_imputed)
```

```
print("\nAfter imputation – Missing Values:", pd.DataFrame(X_imputed).isnull().sum().sum())
```

```
print("Features normalized using StandardScaler ✓")
```

```

# -----

# 3. PCA – Explained Variance

# -----

pca_full = PCA()

pca_full.fit(X_scaled)


cumvar = np.cumsum(pca_full.explained_variance_ratio_)

optimal_k = np.argmax(cumvar >= 0.90) + 1

print(f"\nOptimal components (≥90% variance): {optimal_k}")


# — Plot 1: Scree / Explained Variance —

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

fig.suptitle("PCA – Explained Variance Analysis\n"
"Saumil Mishra | SAP: 500121132 | Batch 2 CCVT",
fontsize=12, fontweight='bold')


comp = range(1, len(pca_full.explained_variance_ratio_)+1)
axes[0].bar(comp, pca_full.explained_variance_ratio_*100,
color='steelblue', edgecolor='white', alpha=0.85)
axes[0].set_xlabel("Principal Component")
axes[0].set_ylabel("Explained Variance (%)")
axes[0].set_title("Scree Plot")
axes[0].grid(axis='y', linestyle='--', alpha=0.5)


axes[1].plot(comp, cumvar*100, 'o-', color='coral', linewidth=2)
axes[1].axhline(90, color='green', linestyle='--', label='90% threshold')
axes[1].axvline(optimal_k, color='purple', linestyle='--',
label=f'Optimal k={optimal_k}')
axes[1].set_xlabel("Number of Components")
axes[1].set_ylabel("Cumulative Variance (%)")

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axes[1].set_title("Cumulative Explained Variance")
axes[1].legend(); axes[1].grid(linestyle='--', alpha=0.5)

plt.tight_layout()
plt.savefig('/home/claude/q1_explained_variance.png', dpi=150, bbox_inches='tight')
plt.close()
print("Saved: q1_explained_variance.png")

# — Plot 2: 2D Scatter —————
pca_2d = PCA(n_components=2)
X_2d = pca_2d.fit_transform(X_scaled)

kmeans_labels = pd.cut(df['Spending_Score'].fillna(50),
bins=3, labels=['Low','Medium','High'])

fig, ax = plt.subplots(figsize=(9, 6))
colors = {'Low': '#e74c3c', 'Medium': '#f39c12', 'High': '#27ae60'}
for label, color in colors.items():
    mask = kmeans_labels == label
    ax.scatter(X_2d[mask, 0], X_2d[mask, 1],
c=color, label=label, alpha=0.7, s=50, edgecolors='white')
ax.set_xlabel(f"PC1 ({pca_2d.explained_variance_ratio_[0]*100:.1f}% var)", fontsize=11)
ax.set_ylabel(f"PC2 ({pca_2d.explained_variance_ratio_[1]*100:.1f}% var)", fontsize=11)
ax.set_title("PCA – 2D Projection of Customer Data\n"
"Saumil Mishra | SAP: 500121132 | Batch 2 CCVT",
fontweight='bold')
ax.legend(title="Spending Level"); ax.grid(linestyle='--', alpha=0.4)
plt.tight_layout()
plt.savefig('/home/claude/q1_pca_2d.png', dpi=150, bbox_inches='tight')
plt.close()
print("Saved: q1_pca_2d.png")

```

```
# — Plot 3: 3D Scatter —————

from mpl_toolkits.mplot3d import Axes3D

pca_3d = PCA(n_components=3)

X_3d = pca_3d.fit_transform(X_scaled)


fig = plt.figure(figsize=(9, 7))

ax3 = fig.add_subplot(111, projection='3d')

for label, color in colors.items():

    mask = kmeans_labels == label

    ax3.scatter(X_3d[mask,0], X_3d[mask,1], X_3d[mask,2],

               c=color, label=label, alpha=0.65, s=40)

    ax3.set_xlabel(f"PC1 ({pca_3d.explained_variance_ratio_[0]*100:.1f}%)")
    ax3.set_ylabel(f"PC2 ({pca_3d.explained_variance_ratio_[1]*100:.1f}%)")
    ax3.set_zlabel(f"PC3 ({pca_3d.explained_variance_ratio_[2]*100:.1f}%)")
    ax3.set_title("PCA – 3D Projection\nSaumil Mishra | SAP: 500121132",
                  fontweight='bold')
    ax3.legend(title="Spending Level")

plt.tight_layout()

plt.savefig('/home/claude/q1_pca_3d.png', dpi=150, bbox_inches='tight')

plt.close()

print("Saved: q1_pca_3d.png")
```

```
# — Plot 4: Biplot (loading vectors) ———

pca_bi = PCA(n_components=2)

Xb = pca_bi.fit_transform(X_scaled)

loadings = pca_bi.components_.T


fig, ax = plt.subplots(figsize=(9,7))

ax.scatter(Xb[:,0], Xb[:,1], alpha=0.3, s=25, color='lightsteelblue')

for i, feat in enumerate(features):
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ax.arrow(0, 0, loadings[i,0]*3, loadings[i,1]*3,
head_width=0.08, head_length=0.05, color='crimson', alpha=0.8)
ax.text(loadings[i,0]*3.2, loadings[i,1]*3.2, feat, fontsize=8, color='darkred')
ax.set_xlabel(f"PC1 ({pca_bi.explained_variance_ratio_[0]*100:.1f}%)")
ax.set_ylabel(f"PC2 ({pca_bi.explained_variance_ratio_[1]*100:.1f}%)")
ax.set_title("PCA Biplot – Feature Loadings\nSaumil Mishra | SAP: 500121132 | Batch 2 CCVT",
fontweight='bold')
ax.grid(linestyle='--', alpha=0.3)
plt.tight_layout()
plt.savefig('/home/claude/q1_biplot.png', dpi=150, bbox_inches='tight')
plt.close()
print("Saved: q1_biplot.png")

```

```

# — Summary —————
print("\n— PCA Summary —————")
for i, ratio in enumerate(pca_full.explained_variance_ratio_[:5], 1):
    print(f" PC{i}: {ratio*100:.2f}% | Cumulative: {cumvar[i-1]*100:.2f}%")
print(f"\n {optimal_k} components explain ≥90% of variance.")
print("—————")

```